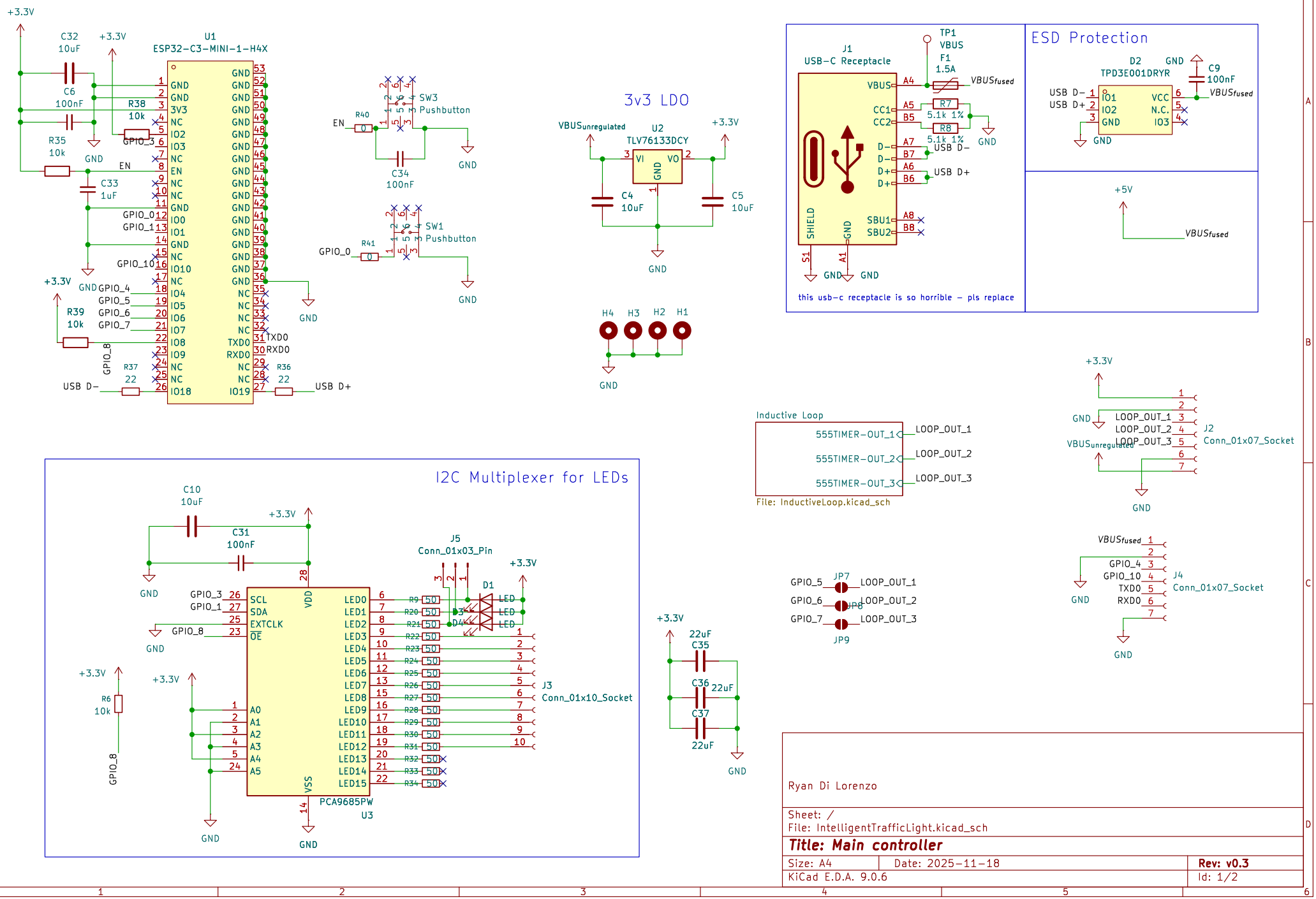
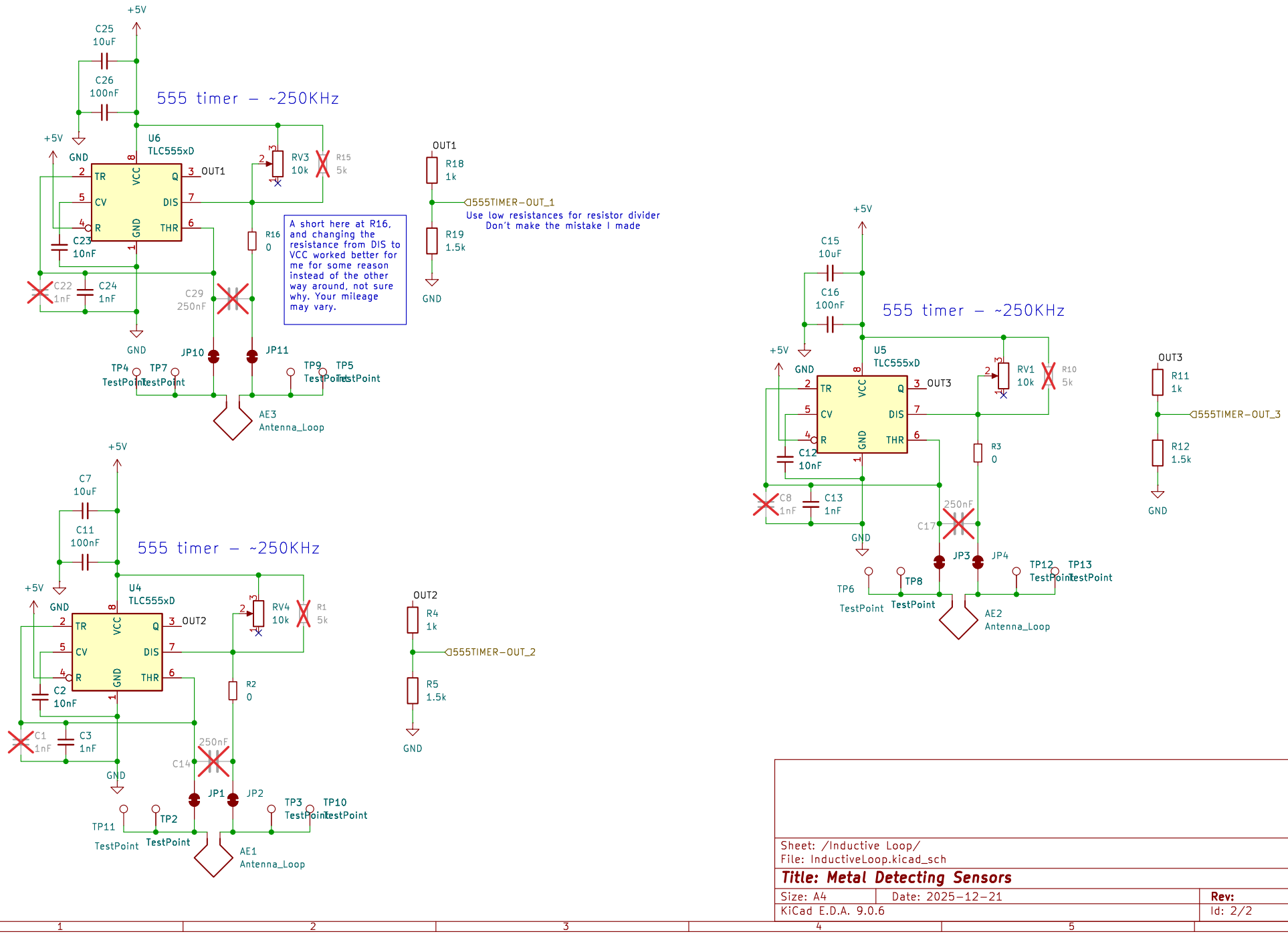

U1
ESP32-C3-MINI-1-H4X
C32
10uF
C6
100nF
R38
10k
R35
10k
C33
1uF
R39
10k
R37
22
R36
22
SW3
Pushbutton
SW1
Pushbutton
R40
R41
C34
100nF
3v3 LDO
U2
TLV76133DCY
C4
10uF
C5
10uF
H4 H3 H2 H1
J1
USB-C Receptacle
TP1
VBUS
F1
1.5A
R7
5.1k 1%
R8
5.1k 1%
this usb-c receptacle is so horrible - pls replace
ESD Protection
D2
TPD3E001DRYR
C9
100nF
Inductive Loop
555TIMER-OUT_1
555TIMER-OUT_2
555TIMER-OUT_3
LOOP_OUT_1
LOOP_OUT_2
LOOP_OUT_3
File: InductiveLoop.kicad_sch
J2
Conn_01x07_Socket
J4
Conn_01x07_Socket
JP7
JP8
JP9
I2C Multiplexer for LEDs
C10
10uF
C31
100nF
J5
Conn_01x03_Pin
D1
R6
10k
PCA9685PW
U3
J3
Conn_01x10_Socket
C35
22uF
C36
22uF
C37
22uF
Ryan Di Lorenzo
Sheet: /
File: IntelligentTrafficLight.kicad_sch
Title: Main controller
Size: A4
Date: 2025-11-18
KiCad E.D.A. 9.0.6
Rev: v0.3
Id: 1/2


555 timer – ~250KHz
U6
TLC555xD
C25
10uF
C26
100nF
C23
10nF
C22
1nF
C24
1nF
C29
250nF
R16
0
RV3
10k
R15
5k
OUT1
R18
1k
R19
1.5k
555TIMER–OUT_1
Use low resistances for resistor divider
Don't make the mistake I made
A short here at R16, and changing the resistance from DIS to VCC worked better for me for some reason instead of the other way around, not sure why. Your mileage may vary.
JP10
JP11
TP4
TP7
TP9
TP5
AE3
Antenna_Loop
555 timer – ~250KHz
U4
TLC555xD
C7
10uF
C11
100nF
C2
10nF
C1
1nF
C3
1nF
C14
250nF
R2
0
RV4
10k
R1
5k
OUT2
R4
1k
R5
1.5k
555TIMER–OUT_2
JP1
JP2
TP11
TP2
TP3
TP10
AE1
Antenna_Loop
555 timer – ~250KHz
U5
TLC555xD
C15
10uF
C16
100nF
C12
10nF
C8
1nF
C13
1nF
C17
250nF
R3
0
RV1
10k
R10
5k
OUT3
R11
1k
R12
1.5k
555TIMER–OUT_3
JP3
JP4
TP6
TP8
TP12
TP13
AE2
Antenna_Loop
Sheet: /Inductive Loop/
File: InductiveLoop.kicad_sch
Title: Metal Detecting Sensors
Size: A4
Date: 2025-12-21
Rev:
KiCad E.D.A. 9.0.6
Id: 2/2